

P3

100 Points Possible

11/12/2025

Attempt 1



In Progress

NEXT UP: Submit Assignment

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Unlimited Attempts Allowed

8/25/2025 to 11/14/2025

Details**Google Page Rank**

In the third paper, you will explore how the Google PageRank algorithm works. A detailed description of this project is given in your book at the end of Chapter 4. This is a guided project, and you will follow the exercises given below:

- Finish writing the hyperlink matrix H from Figure 4.3.
- Write code to implement the iterative process defined previously. Then, create a plot showing how the rank evolves over the iterations.
- What must be true about a collection of n pages such that an $n \times n$ hyperlink matrix H is a stochastic matrix?
- The statement of Theorem 4.9 is incomplete, but the proof is given to you. Fill in the blank in the statement of the theorem, and provide a few sentences supporting your answer.
- Discuss how Theorem 4.9 greatly simplifies the PageRank iterative process described previously. In other words, there is no reason to iterate at all. Instead, find ... what?
- Use the previous two problems to find the resulting PageRank vector from the web in Figure 4.3. Be sure to rank the pages in order of importance.
- Consider the web in Figure 4.4.
 - Write the H matrix and find the initial state x_0 .
 - Find steady state PageRank vector using the two methods described: one using the iterative difference equation and the other using Theorem 4.9 and the dominant eigenvector.
 - Rank the pages in order of importance.

Research other references to the Google PageRank algorithm. After you complete your investigations, you will write a research paper. The tone of your paper is critical to us. Browse papers in the Astrophysical Journal or Physical Review to get a feel for the writing style that we are expecting. You will be submitting a single paper for your group. We have enclosed the LaTeX template that you have to use. Here is the outline of your paper:

- Title
- Authors' Names

- Affiliation Name (The University of Texas at Austin)
- Abstract
- Keywords
- Introduction—In your introduction, you will research the background of this topic and summarize the importance of this algorithm in surfing the web. We want references to papers in published journals, not blogs on the Internet.
- You will state the algorithm and the code that you developed.
- You will present and discuss your results.
- Conclusion - You will conclude by reflecting on how you can improve (if at all) the page rank of your website. You will discuss any future work you could have done by adding randomness to the algorithm Ex 4.100 (book) or EX 4.102 (online).
- Acknowledgment - You will acknowledge any help from your classmates or the TAs.
- References

✓ **View Rubric**

P3 Rubric			
Criteria	Ratings		Points
Title view longer description	Full Marks 4 pts	No Marks 0 pts	/4 pts
Author view longer description	Full Marks 2 pts	No Marks 0 pts	/2 pts
Affiliation view longer description	Full Marks 2 pts	No Marks 0 pts	/2 pts
Abstract	Full Marks 10 pts	No Marks 0 pts	/10 pts
Key	Full Marks 2 pts	No Marks 0 pts	/2 pts
Introduction view longer description	Full Marks 10 pts	No Marks 0 pts	/10 pts
Hyperlink Matrix H view longer description	Exceeds 5 pts	No Evidence 0 pts	/5 pts

Criteria	Ratings		Points
Iterative Process Plot view longer description	Exceeds 5 pts	No Evidence 0 pts	/5 pts
Stochastic Matrix Question view longer description	Exceeds 4 pts	No Evidence 0 pts	/4 pts
Theorem 4.9 Completion view longer description	Exceeds 5 pts	No Evidence 0 pts	/5 pts
Theorem 4.9 Importance view longer description	Exceeds 6 pts	No Evidence 0 pts	/6 pts
Complete PageRank Vector view longer description	Exceeds 4 pts	No Evidence 0 pts	/4 pts
Figure 4.4 - H Matrix view longer description	Exceeds 3 pts	No Evidence 0 pts	/3 pts
Figure 4.4 - Steady State PageRank vector view longer description	Exceeds 3 pts	No Evidence 0 pts	/3 pts
Figure 4.4 - Ranking view longer description	Exceeds 4 pts	No Evidence 0 pts	/4 pts
Algorithm Summary view longer description	Exceeds Summarize you Algorithm/Code 10 pts	No Evidence 0 pts	/10 pts
Conclusion view longer description	Full Marks 8 pts	No Marks 0 pts	/8 pts

Criteria	Ratings		Points
Results Summary view longer description	Exceeds 8 pts	No Evidence 0 pts	/8 pts
Acknowledgements/References	Full Marks 5 pts	No Marks 0 pts	/5 pts

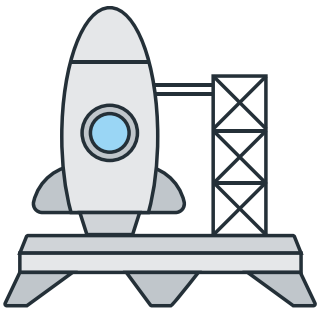
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